Course Code: CSE2711	**Course Name:** Data Structures and Algorithms	
Credits: 3 (2L-0T-2P)	**Contact Hours:** Total 4, 2-hour Theory & 2-hours lab per week	
Batch: 2024, 3rd Sem **Academic Year:** 2025-26	**Semester Duration:** 29 July 2025 to 28 Nov 2025	
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Aim of the course: Data Structures and Algorithms are central to computing. During a computation, programs often store, query, update, and retrieve large volumes of data. There are different ways to design a program to do this information processing - some of which are efficient ways, and others are not. The objective of this course is to teach about the different forms of processing this information and to learn the distinction between the good and the wrong from the perspective of usability and efficiency.

Course Overview and Context: A computer is a data processing device. Storing and retrieving data efficiently are the key concepts for writing computer programs for solving problems. Every computer science student should know these core concepts.
This course focuses on introducing data structures required for storing and organizing data in computer memory and retrieving data efficiently during processing by the computer.
The course covers many commonly used Data Structures like arrays and matrices, linked lists, stacks, queues, trees, graphs, and dictionaries. Algorithms for sorting and searching will also be discussed, emphasizing the analysis of time and space complexities.

Course Outcomes (CO): At the end of the course the students should be able to do the following:

At the end of the course, the students should be able to:

CO1: Understand the core concepts of data representation with suitable data structures and efficient access strategies for solving computational problems.

CO2: Apply the knowledge of data structures and algorithms to create effective solutions to real-world problems keeping the time and space complexities in perspective.

CO3: Analyze a proposed algorithmic solution and identify the constraints and practical limitations posed by the implementation technology.

Prerequisite: if any

1. Mathematics, Programming Fundamentals
2. Object-Oriented Programming

Topics of the course:

Topics	Number of sessions	Course Outcomes
Course overview, introduction to the data structure, Motivation, and need for data structures.	2	CO1
Understanding the role of data structures for efficient data processing, defining data structures, and classification of data structures	2	CO1
A brief idea of algorithms, Asymptotic notations, the notion of time and space complexity	3	CO1
1-D and 2-D arrays, Multi-Dimensional Array, Row-major and column-major ordering	3	CO1
Searching algorithms – Linear, Binary	4	CO2
Sorting algorithms: Bubble, Selection, Insertion, Merge, Quick.	5	CO2
Comparison-based sorting algorithms. Performance analysis of sorting algorithms.	3	CO2
Conceptual understanding of Stack as a data structure and various operations. Applications - Recursion - Tower of Hanoi	4	CO2
Conceptual understanding of Queue as a data structure, various operations. Applications.	4	CO2
Concept of Linked List - Singly, Doubly, Circular, doubly circular linked list; Various operations on the linked list, Applications	3	CO3
Binary Tree and Binary Search Tree: representation using array and linked list; operations – insert, delete, tree traversal	3	CO3
Binary Heap, Heap sort. Applications	3	CO3
Balanced Binary Search Trees - AVL Tree, Operations. Applications	3	CO3
Graphs: Basic terminologies, Traversal Algorithm - BFS, DFS, Spanning Tree	3	CO3
Hashing techniques	3	CO3

CO/PO Mapping:

CO/PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1		2	2					2	1	2		2	2			
CO2	3	3	2					2	1	2		2	3			
CO3			3				3	2	1	2		2	3			

Learning Resources:

Textbook:

1. Data Structures Using C; Aaron M. Tenenbaum, Y. Langsam and M. J. Augenstein; Pearson Education; Ed. First, 2019.
2. Fundamentals of Data Structures in C, Horowitz, Sahni, and Anderson-Freed- University press, Ed. Second, 2008.

Reference Books:

3. Data Structures and Program Design in C; Robert L. Kruse, Bruce P. Leung, Clovis L. Tondo and Shashi Mogalla; Pearson, Ed. Second, 2006.
4. Data Structures with C, Seymour Lipschutz, Schaum's Outline Series, McGraw Hill Education, Ed. First 2017.

Video Links:

1. http://www.nptelvideos.in/2012/11/data-structures-and-algorithms.html
2. https://nptel.ac.in/courses/106/105/106105085/

Note: Instructors will regularly post the necessary learning resources, such as lecture resources, to the online course management portal, i.e., Maitri/Google Classroom.

Experiential Learning Component (50%):

Data structure and algorithms help understand the nature of the problem at a deeper level and, thereby, a better understanding of the world. In this course, students try to comprehend and put into practice something new that hasn't been done before in literature as part of a lab experiment, either by suggesting novel improvements to an existing method, applying existing techniques to new types of data, or proposing a new task or demonstrating the existing with a full explanation. This attempts to provide the following skills:

- Working on data structures and analyzing effects on the performance of algorithms.
- Application of these novel concepts (in whatever form they take)
- Selection of an efficient Data Structure based on a real-world scenario.

Assessment Pattern: The final grade will be based on the marks/ grades obtained in the mid-semester and end-semester evaluation, along with other assessments defined in the assessment table below. The relative grading method described in the university's academic regulations will be followed to grade the students. Every student must secure a minimum of 40% marks after completing all the assessments in the following table to become eligible for grading.

Evaluation Component	Weightage (%)	Evaluation Schedule/Timeline	Rubrics/Remarks
Quiz 1 & 2	2x15% = 30%	Last week of September (15%) and Fourth week of October (15%)	Continuous evaluation – Understanding of theoretical concepts related to Data Structures & Algorithms – Logical, Mathematical and Programming concepts
Lab Assignment 1 & 2	2x15% = 30%	Second week of October (15%) and Third week of November (15%)	– Lab Experiment (Continuous evaluation). Implementation of the data structures using C language and Problem solving. (15%) – Viva for conceptual understanding (15%)
End-Semester Exam	40%	As scheduled by the University	Closed Book, written examination

- All evaluations will be based on the work presented by the students as well as the questions asked, or the problems given to code.
- Cases of AI-generated code or plagiarism will be taken seriously and reported according to the university's policy on Unfair Means (UFM). It is essential that all work is original and adheres to academic integrity.
- There is a mandatory requirement to upload the project to a public repository on GitHub.

Student Responsibilities:

- Attend lectures and do the work Lab Assignments as per instructions.
- Participate in the discussions/assignments held during classes.
- Check announcements on the LMS and emails regularly.
- Submit the assigned task on time.
- Regularly check marks on the LMS to ensure they are up to date.
- Participate in class and take necessary actions to grasp the material. Asking questions is encouraged.
- Communicate any concerns by speaking directly with the instructor.

Attendance Policy: Students are expected to attend classes regularly. Failure to follow the classes regularly and adhere to the expected attendance percentage will result in losing quiz/lab marks and a reduction of the grade as per the University's grading policy.

Recourse Examination Policy: In case a student fails the course, a one-time recourse is permitted as per the academic regulations of the University. Recourse is allowed only for the End Semester examination.

Make-up policy: No make-up exam will be conducted for unexcused absences. The faculty needs to be informed in advance in case the student is not going to appear for any evaluation component, and it is at the discretion of the faculty to sanction makeup for an evaluation component.

Behavior Expectations: No mobile phones and other distractive gadgets are permitted in the class.

Academic Dishonesty/Cheating/Plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.